

Artist-Name Responses beyond a Shared Painting Effect in Text-to-Image Generation

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Abstract

A painter’s name can change images without reproducing painter differences. Six generators rendered 14 fixed scenes twice with artist-free, generic oil-painting, and Monet, Sisley, Pissarro or Cézanne oil-painting prompts, producing 1,008 images. We compare 31 color, spatial and texture measurements from generated images and 649 reproductions. Comparing painters with their four-painter average separates shared change from painter differences. After averaging scenes and correcting repeat noise, 82.5–95.7% of squared change from artist-free to named prompts is shared across names, including the oil-painting clause’s contribution. Before calibration, every model aligns with reference differences; none is shown to beat a predictor making no painter distinctions. Retrospective diagnostics find near-zero or reversed Monet–Sisley alignment in five models. Calibration rescales measured painter contrasts with factors fitted on other scenes. FLUX.2 Max has the lowest uncalibrated error estimate; GPT Image 2 has the lowest held-out calibrated estimate, without changing images. These measurements do not establish perceptual fidelity.

1 Introduction

An instruction to paint “in the style of Monet” can introduce a common painting appearance, exaggerate familiar color or texture cues, or alter the features that distinguish painters. These effects are easy to conflate. Moving closer to Monet’s collection, for example, does not show that a generator reproduces the differences between Monet and Sisley: a generic painting instruction could improve proximity to both.

We ask whether painter names induce differences aligned with those among *documented digital reproductions*, beyond a response shared across names. For each painter, we compare its average color, spatial and texture measurements with the four-painter average, then ask whether generated images reproduce those labeled differences. The target belongs to this reference collection; historical subject mixtures and reproduction processes can contribute to its artist differences. Holding the generated scene text fixed controls the prompt intervention; it does not make those historical subjects identical.

The distinction matters for evaluation. A generator can produce easily separable named conditions whose labels map to the wrong reference painters. It can also match changes along the reference pattern while adding large differences that do not follow it. Conversely, a restrained response may have lower squared error than a stronger response, without distinguishing artists more accurately in every respect. A single proximity score or recognition accuracy cannot separate these possibilities.

Six model configurations receive the same scenes and prompt clauses. Repeated outputs estimate disagreement without counting random output variation as systematic error, under the stated independence assumptions. Separate post-result checks examine painter-pair differences, contrast magnitude, scene dependence and reference quality.

2 Related work

Somepalli et al. (2024) develop learned style descriptors and study similarity across diffusion models. Su et al. (2025) evaluate prompted-artist recognition using content-controlled name substitutions and compare named and artist-free images with real-artist prototypes in CLIP space. Our question concerns the labeled differences among reference painters after removing a shared response; neither name substitution nor cross-model comparison is new.

Deliège et al. (2025) assess stylistic imitation and stereotyping with experts. AI-Pastiche studies authenticity judgments and prompt adherence (Asperti et al., 2025), while AI-WikiArt uses painting-derived captions and vision-language models for attribution (Fu et al., 2025). These tasks distinguish convincing imitation, recognizable authorship and distributional agreement. Here, the primary target is agreement among labeled painter-mean contrasts in measured color, spatial and texture coordinates.

Quantitative art research studies color, composition and image statistics (Kim et al., 2014; Sigaki et al., 2018; Lee et al., 2020). Kim et al. (2026) combine formal and contextual representations for art-historical analysis and contextual image-to-image experiments. Our text-only name intervention supplies no original composition.

Generative evaluation separates fidelity and diversity (Naeem et al., 2020), while conditional metrics distinguish within-condition variation from relationships among means (Benny et al., 2021). We use independent-partition inner products as the primary noise-corrected squared-difference estimator (Diedrichsen and Kriegeskorte, 2017) and energy statistics as a complementary distributional measure (Székely and Rizzo, 2013). Processing sensitivity (Parmar et al., 2022) and limits of learned-embedding separation (Asperti, 2026) motivate the source-quality checks and the restriction to a specified digital reference target.

3 Design and measurements

3.1 Painters and reference images

The primary panel covers four related painters through 649 distinct reproductions: 297 Monet, 106 Sisley, 141 Pissarro and 105 Cézanne. Artist means receive equal weight, so the larger Monet collection does not dominate the between-artist target. A separate 221-work development panel sets common feature units; no primary reference or generated image is used to fit that transformation. Both panels were assembled before the present experiment and had been used in earlier analyses. They are explicit, finite comparison panels rather than newly withheld samples of each painter’s complete oeuvre.

Reference files carry Wikimedia Commons source metadata and Wikidata work identities (Wikimedia Commons, 2026; Vrandečić and Krötzsch, 2014). These support traceable selection and rights records, but do not standardize photography, color calibration or conservation state. These are measurements of digital reproductions, whose capture conditions can affect the comparison.

3.2 A common-scene, six-model experiment

We compare GPT Image 1, GPT Image 2, GPT Image 2.5 Flare (Flare), GPT Image 2.5 Sunburst (Sunburst), Nano Banana 2 and FLUX.2 Max (FLUX). Fourteen fixed outdoor scene descriptions (briefs) span water, built, land and mixed compositions. Each scene is crossed with six clauses: no painting instruction; “Render as an oil painting”; and “Render as an oil painting in the style of [painter]” for each of the four painters. The scene text is otherwise identical. Each model–scene–clause cell receives two separately requested outputs, giving 1,008 images collected on 10 September 2026.

All models receive text only and return 1,024-by-1,024-pixel images. The four OpenAI models use medium quality; the other two use their specified 1K configurations. These comparisons concern the recorded configurations, not maximum attainable quality or equal expenditure per image. Served model versions were not independently verified (Appendix G). The exact prompts, parameters and randomized request order were fixed before collection. The 14 scenes are a fixed design, not a random sample of all possible content. Earlier, separate prompt interventions are summarized in Appendix F.

Figure 1 shows all six conditions for one brief. GPT Image 2’s Cézanne trees have darker blue-green outlines than its Monet and Sisley trees: a visible name response in this scene, whose agreement with historical painter differences remains to be measured.

3.3 What the features measure

The 31 features in Table 1 capture properties that need not change together. Two images can have similar chroma (color strength) but different hue diversity, or similar total contrast but different edge directions and fine-scale texture. Such distinctions make results by feature family interpretable. Correlated coordinates and sensitivity to image processing nevertheless prevent treating Euclidean distance as a validated measure of artistic style.

Family	Measurements	Interpretation
Color (11)	Lightness and chroma median/IQR; chromatic fraction; hue concentration and entropy; color differences at three lags and their slope	Brightness, color strength and diversity, and how rapidly colors change over space.
Spatial (8)	Spectral slope, residual and anisotropy; orientation entropy; horizontal–vertical balance; gradient median/IQR; quadrant divergence	Coarse versus fine organization, dominant directions, edge strength and regional variation in orientation.
Texture (12)	Four wavelet energies, slope and curvature; three local-binary-pattern entropies; three local coefficients of variation	Detail across scales, diversity of local patterns and local luminance contrast.

Table 1: The 31 fixed digital features. Counts refer to scalar coordinates. Texture measurements do not measure physical paint relief or authenticate brushwork. Exact scales and definitions appear in Appendix A.

Images undergo orientation correction and conversion to sRGB using valid embedded profiles; compatible unprofiled files are treated as sRGB. The primary pipeline preserves aspect ratio at a 512-pixel short side, using Lanczos resizing without upsampling. No painting-region mask is applied: source margins or calibration strips can enter the measurements. Each coordinate is centered and scaled by its median and interquartile range in the development panel, with equal total weight per painter. Thus a unit is a development-panel IQR in that coordinate, not a perceptual just-noticeable difference. The Euclidean metric weights standardized coordinates equally, without further equalizing feature families.

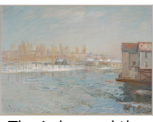
	Monet	Sisley	Pissarro	Cézanne	Artist-free	Generic painting
Reference reproduction	 Rocks at Port-Goulphar	 The Loing and the Mills of Moret	 River Oise near Pontoise	 Along the Banks of the Marne		
GPT Image 1						
GPT Image 2						
GPT Image 2.5 Flare						
GPT Image 2.5 Sunburst						
Nano Banana 2						
FLUX.2 Max						

Figure 1: One scene, six prompt conditions and six generators. The requested scene is a broad river bending past a gravel bank, three trees on the opposite bank, and a low hill under an overcast sky. In each model row, compare the four named outputs with the artist-free and generic oil-painting controls on the right, then compare the painter names with one another. The top row shows audited reference regions for display; primary scores use full frames. These works were neither generator inputs nor matched compositions. All generated examples use the first brief and first repeat; Appendix D gives the full prompt, selection rule and provenance.

4 Comparing painter differences

4.1 What counts as agreement?

The comparison has three steps. Measure the same 31 properties in every image. For each historical painter, subtract the average of all four painters; this records how that painter differs from the group. Repeat the subtraction among the four named outputs for each generated scene, then compare the corresponding painter differences. We call these differences *contrasts*.

Consider median lightness: Monet averages .325 development-IQR units below Sisley in the references. Across all 14 scenes and both repeats, GPT Image 2 has a smaller gap, .254, in the same direction. Subtracting each set’s four-painter mean preserves these gaps; a shared brightness shift cannot create this distinction. The full comparison uses all 31 coordinates and four painters.

Two scores answer different questions. The *aligned response*, β , measures how far generated contrasts extend along the reference contrasts; $\beta = 1$ matches that component’s strength. The *error*, D , counts total disagreement, including differences in other directions. Consider three idealized generators: one makes no painter distinctions, one exactly matches the reference contrasts, and one doubles them. Their expected errors are respectively 1, 0 and 1. A stronger response can therefore have more error, and a near-unit aligned response can coexist with substantial mismatch elsewhere.

4.2 Computing the two scores

Let $\mu_a \in \mathbb{R}^{31}$ be painter a ’s standardized reference mean and $r_a = \mu_a - \frac{1}{4} \sum_b \mu_b$ its contrast. The total squared size of these contrasts, $H = \sum_a \|r_a\|^2$, normalizes both scores. Here $\|u\|^2$ sums squared coordinates, $\langle u, v \rangle$ sums matching-coordinate products, and hats mark estimates. For generated measurements z_{msak} , indices denote model m , scene s , painter a and repeat $k \in \{1, 2\}$:

$$d_{msak} = z_{msak} - \frac{1}{4} \sum_b z_{msbk}. \quad (1)$$

Centering removes shared additive shifts, but shared rescaling or mixing of coordinates can still change the contrasts (Appendix B.1). With $S = 14$ scenes, the aligned response is

$$\hat{\beta}_{msk} = \frac{\sum_a \langle d_{msak}, r_a \rangle}{H}, \quad \hat{\beta}_m = \frac{1}{S} \sum_s \frac{\hat{\beta}_{ms1} + \hat{\beta}_{ms2}}{2}. \quad (2)$$

A positive value indicates an overall response along the reference differences; it does not require agreement for each artist pair. Values above one indicate exaggeration along that direction. To measure error, multiply the discrepancies from the two repeats rather than squaring either repeat’s discrepancy:

$$\hat{D}_{ms} = \frac{1}{H} \sum_a \langle d_{msa1} - r_a, d_{msa2} - r_a \rangle, \quad \hat{D}_m = \frac{1}{S} \sum_s \hat{D}_{ms}. \quad (3)$$

Under stable conditional means and independent, mean-zero repeat errors, noise that does not recur across requests contributes zero in expectation. The expected score is then the normalized squared error of the scene-conditional mean contrasts. Individual estimates can be negative; they are retained, not interpreted as better-than-perfect recovery. The $D = 1$ no-contrast benchmark is a theoretical predictor, not the observed generic-painting arm.

The target is the same pooled historical contrast in every scene. A model whose artist differences vary across scenes can therefore incur error even if its across-scene average matches the references. Such variation need not be an artistic mistake. We separately report mismatch after averaging scenes and the added scene-variation component; their sum is D . Exact identities and derivations are in Appendix B.1.

4.3 Model comparisons and uncertainty

We compare models scene by scene, then average their paired error differences. This controls for the common brief; features, painters and individual images are not treated as independent replicates.

Before inspecting new feature outcomes, we declared six aligned responses and 15 model-pair error differences. Paired-scene Student intervals use Bonferroni adjustment for these 21 comparisons together, giving approximate 95% simultaneous coverage. Intervals excluding zero resolve their comparisons. Absolute- D intervals are nominal 95% summaries: they are unadjusted and descriptive.

The intervals reflect variation across the 14 fixed briefs, combining generation noise with real scene differences. They neither isolate fresh-request uncertainty for this exact panel nor establish coverage for unseen scenes. Two outputs enable noise correction but do not establish its sampling distribution or service independence.

Declared checks vary scenes, reference samples, feature families and measurement windows. A separately declared sensitivity equalizes four title-derived subject classes within each painter. These choices probe dependence on the reference panel and metric; they do not match historical compositions. Their full specifications remain in Appendix B.

4.4 What the additional comparisons test

The artist-free and generic-painting arms characterize the common response removed by centering. Retrospective diagnostics address three questions: which painter pairs contribute to the aggregate response, how contrast magnitude changes error, and how reference choices change the comparison. For the magnitude check, we choose one nonnegative multiplier per model, shared across all painter contrasts and coordinates, to minimize error on 13 scenes. We evaluate it on the omitted scene, rotating through all 14 scenes. This *calibration* multiplies only centered generated contrasts; reference contrasts, coordinate units and images stay fixed. These diagnostics interpret the declared results without extending their significance-test family. Formulas appear in Appendix B.1; controls, fits and the source audit are in Appendix C.

5 Results

5.1 A response in the right direction can still miss the target

All 1,008 outputs yield valid measurements. With the full-frame reference target, every model’s aligned-response interval lies above zero (Table 2). A purely common additive response is therefore insufficient to explain the named conditions under the stated inference model. Yet positive alignment does not ensure low error. GPT Image 2 nearly matches the reference-aligned strength, $\beta = .999$ [.893, 1.104], while its total error is $D = 1.226$: it adds substantial differences beyond that aligned component.

FLUX.2 Max has the lowest error estimate, .801, despite a smaller aligned response, .470. Adjusted comparisons resolve higher error for Flare and Sunburst than for FLUX; the other 13 intervals include zero (Appendix E, Table 10). These use full-frame references; the Sunburst result changes after source correction (Section 5.5). No model is established to beat the no-contrast benchmark of one. Thus the study detects a reference-aligned response more clearly than it establishes low total mismatch.

In Figure 2, Cézanne remains visibly separated, while generated Monet and Sisley means often lie close together. The M/S/P detail boxes magnify the nearby positions. The axes combine the 31 standardized features, with directions fitted only to the four reference means. Together they show 86.99% of variation among those means; the pair analysis below checks all 31 features.

Model	Aligned response β	Uncalibrated error D	ΔD versus FLUX
GPT Image 1	0.938 [0.709, 1.168]	1.572	0.771 [-0.037, 1.580]
GPT Image 2	0.999 [0.893, 1.104]	1.226	0.425 [-0.161, 1.011]
GPT Image 2.5 Flare	0.772 [0.680, 0.864]	1.738	0.937 [0.256, 1.618]
GPT Image 2.5 Sunburst	0.824 [0.680, 0.968]	1.641	0.840 [0.058, 1.623]
Nano Banana 2	0.442 [0.256, 0.629]	1.109	0.309 [-0.847, 1.464]
FLUX.2 Max	0.470 [0.239, 0.701]	0.801	0 (baseline)

Table 2: Agreement with the full-frame reference target. $\beta = 1$ matches its aligned amplitude; larger is not necessarily better. Expected D is 0 for exact agreement and 1 for no painter distinctions. Positive ΔD favors FLUX.2 Max. Table brackets give approximate 95% simultaneous intervals from the prespecified family of six aligned responses and 15 scene-paired model contrasts. FLUX’s lowest D estimate has an unadjusted 95% interval [0.586, 1.016], spanning the no-distinction value 1; this interval is descriptive.

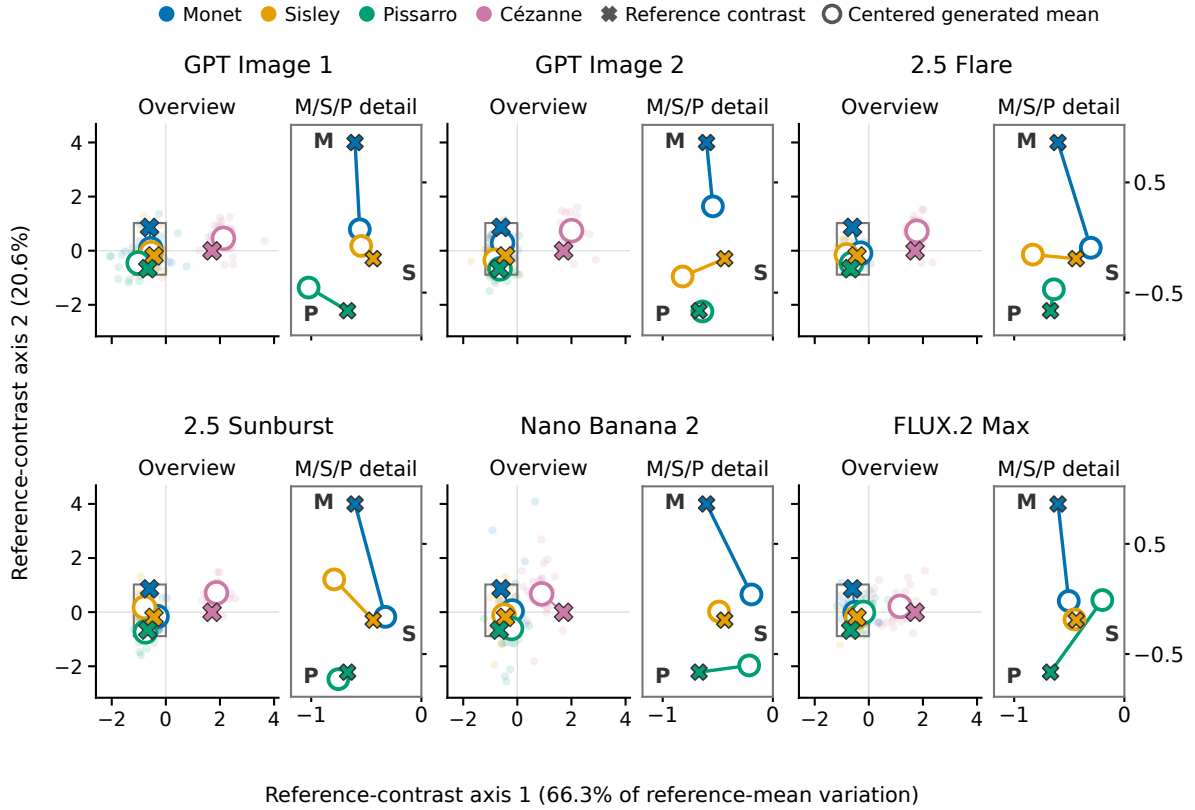


Figure 2: Matching the labeled painter pattern. Crosses mark reference contrasts; open circles average generated contrasts over all 14 scenes and both repeats. Shorter matching-label lines indicate less projected mismatch; faint dots show centered scene/repeat contrasts. The outlined region is enlarged in the adjacent Monet (M), Sisley (S) and Pissarro (P) detail at a common scale; Cézanne remains in the overview. Each view uses common axes across models. Agreement in these two projected axes does not establish agreement in all 31 features.

5.2 Most scene-averaged change is common to the painter names

After averaging scenes, named-minus-artist-free shifts comprise a four-name mean and painter-specific departures. The mean includes the oil-painting clause and any common naming response: 82.5–95.7% of repeat-corrected squared change (Table 3). Averaging emphasizes consistent shifts; splitting within scenes retains scene-specific variation and gives 88.6–97.2% (Appendix B). β measures departures along the reference contrasts.

	GPT Image 1	GPT Image 2	Flare	Sunburst	Nano Banana 2	FLUX.2 Max
Shared (%)	82.5	85.8	84.8	83.9	88.1	95.7
Between-name (%)	17.5	14.2	15.2	16.1	11.9	4.3
Corrected cosine	0.689	0.774	0.766	0.837	0.930	0.916
Squared-size ratio	1.986	1.960	2.816	3.356	3.692	4.138

Table 3: Shares of scene-averaged, repeat-corrected squared change from artist-free to named prompts, including the painting clause. Shared is the four-name mean shift; between-name is departure from it. The lower rows give retrospective comparisons of the shared named shift with the generic oil-painting shift, using the same artist-free baseline. The noise-corrected cosine measures their directional agreement (1 means parallel); the ratio divides their repeat-corrected squared magnitudes, shared by generic.

The shared named shift points broadly along the generic oil-painting shift (noise-corrected cosine .689–.930; 1 means parallel), but its repeat-corrected squared magnitude is 1.96–4.14 times as large. Both use scene averages and the same artist-free baseline; Table 3 lists each model’s values. Thus, a response shared across painter names need not be identical to the response to a generic painting instruction. The generic shift and the additional shared naming shift can reinforce or oppose one another, so their squared magnitudes do not add as independent contributions (Appendix B).

5.3 The aggregate response conceals uneven painter distinctions

Retrospective pair diagnostics extend the lightness example in Section 4.1 to all 31 features (Figure 3). Monet–Sisley aligned responses range from $-.249$ to $.129$ in five models, versus $.402$ for GPT Image 2. Yet GPT Image 1 has lower estimated error for this pair ($.56$ versus 1.16): stronger alignment does not ensure lower mismatch. Omitting Cézanne lowers FLUX’s overall response from $.470$ to $.096$ and GPT Image 2’s from $.999$ to $.651$.

Swapping Monet and Sisley lowers error for Flare, Sunburst and Nano Banana 2. Reversing a negatively aligned pair difference lowers error without changing its size.

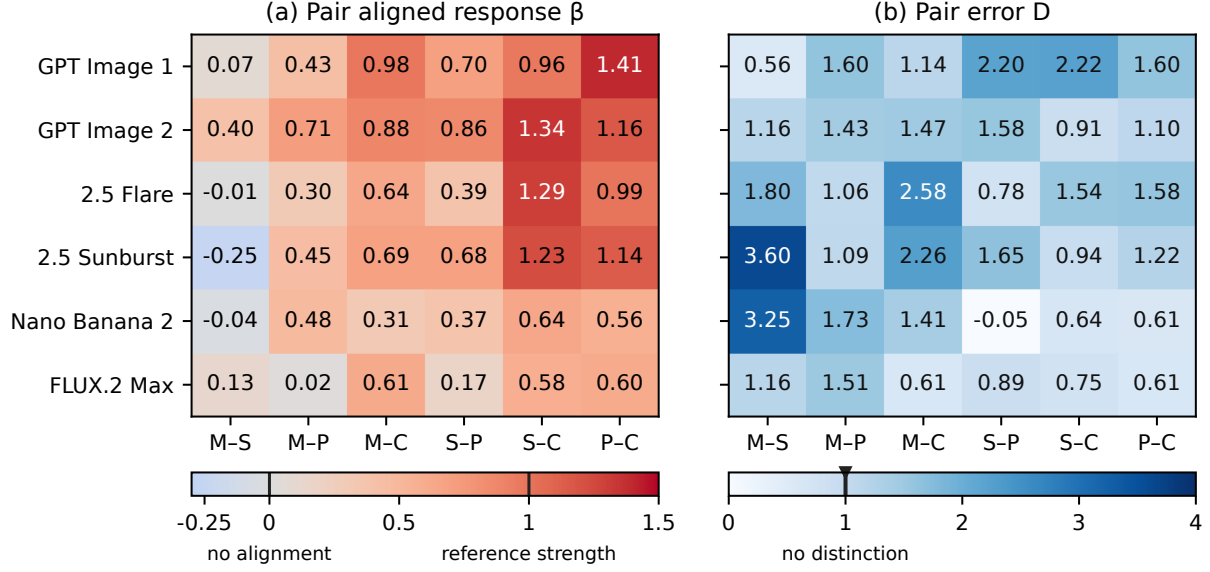


Figure 3: Descriptive painter-pair estimates (no pairwise tests). M, S, P and C denote Monet, Sisley, Pissarro and Cézanne. Pair aligned response 0 means no aligned component, 1 matches reference strength, values above 1 exaggerate it, and negative values indicate the opposite direction. Expected error is 0 for exact agreement and 1 for no distinction. Each pair uses all 31 features and is normalized by its squared reference distance. Negative error can arise from repeat correction, not better-than-exact agreement.

5.4 Scene averaging and contrast magnitude change the ordering

We compare fit scene by scene, after averaging scenes, and after held-out rescaling of painter contrasts (Table 4).

Averaging scenes. Nano Banana 2’s mean painter contrasts have lower error than FLUX’s (.723 versus .761). Evaluating each scene adds a larger variation component for Nano Banana 2 (.387 versus .040), reversing that order. D penalizes variation around the pooled historical target; averaging can cancel scene-dependent differences.

Model	Uncalibrated scene error D	Error after averaging scenes $D_{\text{aggregate}}$	Scene variation V_{scene}	Calibrated scene error D_{held}
GPT Image 1	1.572	1.239	0.333	0.645
GPT Image 2	1.226	1.044	0.182	0.554
2.5 Flare	1.738	1.483	0.255	0.741
2.5 Sunburst	1.641	1.337	0.305	0.706
Nano Banana 2	1.109	0.723	0.387	0.832
FLUX.2 Max	0.801	0.761	0.040	0.714

Table 4: Retrospective error decomposition and calibration against full-frame references. $D = D_{\text{aggregate}} + V_{\text{scene}}$. Calibration rescales centered generated contrasts using the other 13 scenes and evaluates them on the omitted scene; it does not produce new images. Bold marks each error column’s minimum, not significance.

Rescaling contrasts. GPT Image 2’s aligned response is near one, but its squared contrast size exceeds twice the reference’s ($Q = 2.223$, where 1 matches the reference size; Appendix C.2). Shrinking weakens its reference-aligned component, but reduces off-target differences enough to lower the total error estimate. GPT Image 2’s fitted multipliers scale every contrast coordinate by about 0.45, using only the other 13 scenes each time.

After calibration, GPT Image 2 has the lowest error estimate, .554, compared with FLUX’s .714; it has lower scores in 12 of 14 held-out folds. Deleting any scene and refitting the entire procedure preserves the mean advantage (Appendix C.2). The folds overlap, so their win count does not establish statistical superiority.

5.5 Which comparisons survive reference changes?

For the uncalibrated scene error D , FLUX retains the lowest point estimate across scene deletions, measurement windows, reference targets and feature-weighting sensitivities (Appendices B.2 and C).

Source handling has a more consequential effect on statistical comparisons. AI assistants inspected all reference and development reproductions, selecting crops that exclude peripheral artifacts in 90 reference and 41 development images. The audit used no human verification. After cropping and refitting development scaling, FLUX’s error estimate changes from .801 to .872. Corrected aligned-response intervals remain above zero for all six models, and GPT Image 2 retains the lowest calibrated estimate.

Table 5 makes the changed comparisons explicit. With corrected regions and refitted scaling, the Sunburst interval crosses zero, Flare still has higher error than FLUX and the GPT Image 1 interval moves above zero. The latter is a retrospective sensitivity, not a new confirmatory finding. Source correction preserves FLUX’s lowest point estimate while changing the evidence for particular model differences.

Model comparison	Full-frame difference [adjusted interval]	Crop + scaling correction Difference [adjusted interval]
GPT Image 1 – FLUX	0.771 [−0.037, 1.580]	0.758 [0.026, 1.491]
2.5 Flare – FLUX	0.937 [0.256, 1.618]	0.864 [0.193, 1.535]
2.5 Sunburst – FLUX	0.840 [0.058, 1.623]	0.705 [−0.026, 1.436]

Table 5: Uncalibrated error differences before and after source correction, using audited painting regions and refitted development scaling. Positive differences favor FLUX.2 Max. Full-frame intervals use the 95% simultaneous procedure across 21 endpoints; corrected intervals reuse that procedure retrospectively and add no confirmatory tests.

A separate check uses class-specific reference means for 11 water, built and land briefs, excluding three mixed briefs. Replacing title-derived reference labels with visual labels retains FLUX’s lowest error estimate. Appendix C gives the controls and audit.

6 Discussion

Detecting a name response and matching painter differences are distinct achievements. FLUX’s lowest uncalibrated error estimate coexists with weak Monet–Sisley alignment. Model comparisons should therefore report overall and pairwise scores together for the stated reference target, distinguishing estimates from resolved comparisons. Reports should also state whether they score each scene, the average pattern, or held-out contrasts rescaled using other scenes: these choices change the model ordering.

The intervention identifies differences between prompt conditions, not their internal causes. Training composition, guidance and shared transformations remain possible explanations.

6.1 Scope and remaining limitations

The target is the pattern of differences in a finite digital reference collection. It mixes subject choices and reproduction processes with properties of the paintings. Coarse content classes do not match composition, viewpoint, career period or actual generated scene adherence. The source audit addresses visible peripheral regions and title/visual-class disagreements, but AI judgments, uncertain boundaries and remaining capture differences limit that correction. One reproduction per work cannot separate effects of illumination, sharpening, conservation or digitization source.

The 31 coordinates describe digital color, spatial organization and texture, not physical brushwork or perceived painter resemblance; numerical replay and robustness to coordinate weights do not validate those judgments. Inference also conditions on four related painters, 14 authored briefs, two repeats and one clause template. Simulations probe stated noise models, not actual service independence. The image panels make the comparison inspectable, while incomplete public exact-pixel access still limits independent measurement reproduction.

7 Conclusion

Most squared feature change from artist-free to named prompts is shared across names after scene averaging, including the oil-painting contribution. All six models nevertheless show reference-aligned painter differences, yet none is established to beat the no-distinction error benchmark before calibration. Uneven painter-pair agreement and the changed error ordering after calibration show why a single overall score is insufficient.

Data and code availability

The [repository](#) contains protocols, prompts, request settings, source metadata, scalers and feature vectors. The primary analysis, diagnostic extensions and source-quality sensitivity have separate versioned records with input hashes and replay code. The source audit records all work identities, region boxes, label decisions and corrected vectors; the original results remain intact.

Figure 1 displays 36 generated and four reference examples. Appendix D explains their selection; the accompanying manifest records identities and byte hashes. Public numerical replay begins with vectors; independent re-extraction requires the exact source pixels, which remain outside the compact repository. Earlier experiments have [public releases](#); the current experiment and corrections still need a dedicated public archive with permissible exact pixels or a verified recovery route.

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A Feature definitions

Color coordinates use CIELAB under D65 and the 2° observer. Lightness and chroma each contribute a median and IQR. Chromatic fraction is the share of pixels with $C^* \geq 5$. Hue concentration is the magnitude of the chroma-weighted circular mean; hue entropy uses 24 equally spaced bins and is normalized by their log count. Both use chroma weights and are zero when fewer than 1% of pixels are chromatic. At each of three lags (1%, 4% and 16% of the short side), the median is taken over pooled horizontal and vertical CIEDE2000 differences. A log–log slope summarizes their scale dependence (Sharma et al., 2005).

Spatial and texture coordinates use linear-sRGB luminance. A Tukey window (parameter .1) precedes Fourier analysis. Thirty-two logarithmic radial bins cover 4–128 cycles per short side; a Theil–Sen line gives spectral slope and its root-mean-square residual. Power-weighted axial concentration measures anisotropy. Scharr derivatives supply gradient median and IQR, an 18-bin magnitude-weighted orientation entropy, and horizontal–vertical balance. Mean Jensen–Shannon divergence of quadrant orientation histograms from the global histogram measures regional variation.

Four log mean-squared detail energies come from a four-level undecimated db2 wavelet transform, followed by their linear slope and mean second difference (Mallat, 1989). Uniform local-binary-pattern entropy uses $(P, R) = (8, 1), (16, 2), (32, 4)$ on quantized luminance (Ojala et al., 2002). Three median local coefficients of variation use window widths obtained by rounding 1%, 4% and 16% of the short side and increasing even widths to the next odd integer, with a 10^{-6} mean floor. The fixed implementation records boundary handling and stabilizing constants. The separate scaler contains 101 Monet, 36 Sisley, 48 Pissarro and 36 Cézanne works. Equal painter mass is used to calculate each weighted median and IQR.

B Estimator details and sensitivity analyses

Let $\theta_{msa} = \mathbb{E}[d_{msak}]$ and $d_{msak} = \theta_{msa} + \eta_{msak}$. If repeat errors have zero mean and $\mathbb{E}\langle \eta_{msa1}, \eta_{msa2} \rangle = 0$, then

$$\mathbb{E}[\widehat{D}_{ms}] = H^{-1} \sum_a \|\theta_{msa} - r_a\|^2.$$

Centering induces dependence between artists within a repeat, but independent repeat blocks preserve the vanishing cross-repeat term. A common response has $\theta_{msa} = 0$ and hence expected $D = 1$, even if it shifts the entire named collection toward the references. A correctly aligned but rescaled response $\theta_{msa} = br_a$ gives expected $D = (b - 1)^2$. Orthogonal deviations add their

squared magnitude. This identity explains why stronger painter contrasts need not have lower reference-relative squared error.

The aggregate error $D_{\text{aggregate}}$ applies the cross-product estimator to scene-averaged contrasts; Equation 5 relates it to the primary, equally scene-weighted error. Painters also receive equal weight. Their additive contributions sum to D , but centering couples the four conditions, so these contributions are not standalone artist-fidelity scores.

For a model contrast, compute one error difference per complete paired scene. Its standard error is the sample standard deviation of those differences divided by $\sqrt{n}$. The simultaneous half-width is $t_{n-1, 1-.05/(2 \cdot 21)} \text{SE}$. Six aligned-response intervals use the same family adjustment. Additional nominal intervals and 5,000 paired-scene bootstrap draws are descriptive checks, not alternative routes to significance. Reference sensitivity draws 1,000 bootstrap samples independently within each painter, recomputes reference means and H , and holds generated vectors and scaling fixed. It describes reference-panel dependence rather than all uncertainty jointly.

Declared before inspecting the new feature outcomes, the four-class reference sensitivity uses a fixed lexicon applied to source titles: water-organized, built-place-organized, route-organized, and open or wooded land. Each class mean receives one-quarter weight within a painter. Resampling within artist/class strata conditions on those labels and does not account for classification error. The older controlled naming panel used a separate three-class visual coding scheme, described below.

For the shared-effect diagnostic, average each arm over scenes within repeat. With named-minus-free differences h_{ak} and $c_k = \frac{1}{4} \sum_a h_{ak}$, common and specific squared magnitudes are $4\langle c_1, c_2 \rangle$ and $\sum_a \langle h_{a1} - c_1, h_{a2} - c_2 \rangle$. Their common fraction is reported when the total is positive; corrected components may be negative. The within-scene version computes both squared-change components separately in each scene, averages each over scenes, then takes their ratio. Its shares range from 88.6% (GPT Image 1) to 97.2% (FLUX). Let g_k be the generic-minus-free shift. The ordinary repeat-averaged cosine shares a noisy free estimate in both vectors, adding its error covariance trace to the expected numerator. The corrected cosine reported in Section 5.2 instead uses cross-repeat products, $[\langle g_1, c_2 \rangle + \langle g_2, c_1 \rangle] / [2\sqrt{\langle g_1, g_2 \rangle \langle c_1, c_2 \rangle}]$ when both norm squares are positive. The ratio itself is not unbiased or constrained to $[-1, 1]$. Primary D uses only named arms and is unaffected by shared-free-control noise.

B.1 Additional score identities

Under a shared affine transformation $z \mapsto Az + b$, centering removes the additive term b but leaves the transformed contrast Ad .

The estimator has an exact decomposition. Write $e_{msak} = d_{msak} - \hat{\beta}_{msk} r_a$. Then

$$\hat{D}_{ms} = (\hat{\beta}_{ms1} - 1)(\hat{\beta}_{ms2} - 1) + \frac{1}{H} \sum_a \langle e_{msa1}, e_{msa2} \rangle. \quad (4)$$

The first term measures reference-aligned amplitude error. The residual is orthogonal to the single stacked reference configuration; it is not necessarily outside the feature-space span of the reference means, much less outside all authentic artistic variation. Both terms can be negative in finite samples.

Let θ_s stack the four conditional mean contrasts and $\bar{\theta} = S^{-1} \sum_s \theta_s$. The population target decomposes as

$$D_{\text{scene}} = \frac{\|\bar{\theta} - r\|^2}{H} + \frac{1}{SH} \sum_s \|\theta_s - \bar{\theta}\|^2 = D_{\text{aggregate}} + V_{\text{scene}}. \quad (5)$$

To separate contrast strength from direction, define

$$\hat{Q}_m = \frac{1}{SH} \sum_{s,a} \langle d_{msa1}, d_{msa2} \rangle, \quad \hat{D}_m = 1 - 2\hat{\beta}_m + \hat{Q}_m. \quad (6)$$

For positive $\hat{Q}_m$, $\hat{\beta}_m/\sqrt{\hat{Q}_m}$ is a noise-corrected alignment summary; finite estimates need not lie in $[-1, 1]$. Scaling every centered generated contrast by c gives $\hat{D}_m(c) = 1 - 2c\hat{\beta}_m + c^2\hat{Q}_m$. We fit $c = \max(0, \hat{\beta}/\hat{Q})$ on the other 13 scenes and evaluate it on the held-out scene. The fitting criterion is available only when its training $\hat{Q}$ is positive, as it is in every observed fold. This counterfactual concerns feature vectors, not an implemented image transformation.

B.2 Paired comparisons and declared sensitivities

We fit the scene fixed-effects regression

$$\hat{D}_{ms} = \alpha_s + \gamma_m + \varepsilon_{ms}, \quad \gamma_{\text{GPT Image 2}} = 0. \quad (7)$$

In the balanced panel, coefficient differences equal the paired scene means used for inference. Changing the regression baseline leaves these differences unchanged; Table 2 presents them relative to FLUX. Declared sensitivities delete each scene, resample scenes and reference works, examine feature families and omit texture. Central-square windows control aspect ratio by discarding peripheral content in generated and reference images. FLUX retains the lowest uncalibrated error after every scene deletion ($D = .754-.852$) and with square windows ($D = .765$; next-lowest Nano Banana 2, 1.024).

B.3 Distributional proximity and repeat spread

For painter a , energy discrepancy compares its generated collection Y_a with reference collection X_a :

$$\hat{\mathcal{E}}(X, Y) = \frac{2}{n_X n_Y} \sum_{i,j} \|x_i - y_j\| - \frac{1}{n_X^2} \sum_{i,i'} \|x_i - x_{i'}\| - \frac{1}{n_Y^2} \sum_{j,j'} \|y_j - y_{j'}\|. \quad (8)$$

Its finite-sample expectation depends on within-collection spread; equal generated sample sizes do not eliminate comparison bias. Appendix C gives the correction for the repeated-scene design. Within-scene repeat spread, $\frac{1}{2S} \sum_s \|z_{msa1} - z_{msa2}\|^2$, is the mean sample-variance trace of the two outputs. It separates repeat dispersion from variation across briefs.

C Post-result target diagnostics

These retrospective analyses retain the completed 1,008 generated images and the original inferential family. The initial diagnostics use unchanged reference vectors and scaling; the subsequent source audit separately varies painting regions, development scaling and content labels.

C.1 Content-specific targets and genuine-painting controls

The recorded reference classes are water-organized, built-place-organized, route-organized, and open or wooded land, assigned by a fixed source-title lexicon. Counts in that order are (191, 67, 8, 31) for Monet, (52, 29, 9, 16) for Sisley, (28, 77, 12, 24) for Pissarro and (31, 28, 12, 34) for Cézanne. Class-specific means are centered across painters just as in the pooled target. Their mean squared departure from the pooled target is $.438H$, a description of observed reference means that includes finite-sample error.

For the generated comparison, the four water, four built and three land briefs use the corresponding class means; the three mixed briefs are excluded. We average the class-specific cross-product error over these $S = 11$ scenes and divide by $H_c = S^{-1} \sum_s \sum_a \|r_{a,c(s)}\|^2 = 7.413$, compared with pooled $H = 5.915$. Each column of Table 7 therefore has its own denominator. The unchanged 11-scene pooled comparison isolates the scene-subset choice. These title-derived

strata have 16–191 reference works per artist/class among the three included classes, but do not supply visually matched historical scenes.

For each of 1,000 genuine-painting controls, works in each artist/class cell are randomly divided into disjoint halves, with the smaller half used for target estimation. A pooled control samples two independent works with replacement from each painter’s held-out half for each of 14 pseudo-scenes. A class-stratified control samples within the held-out class for each of the 11 water, built or land pseudo-scenes. It is compared with both pooled and class-specific means from the training halves, using each target’s own squared magnitude. The random seed is 2026091201. Sampling can repeat a held-out work, but no held-out work is used to estimate that split’s target. The development scaler is fixed throughout.

To separate sparse sampling from changes in the comparison pools, we also calculate each split’s expected score from its exact held-out means, keeping the training target and normalizer fixed (Table 6).

Control sampling	Reference target	Two draws	Exact held-out means
Pooled	Pooled	[−.758, 1.380]	[.135, .428]
Class-stratified	Pooled	[−.554, 2.400]	[.544, .993]
Class-stratified	Class-specific	[−.248, 1.946]	[.517, .954]

Table 6: Central 95% ranges of genuine-painting control scores across 1,000 splits. Exact means integrate out sampling within each split; these finite-pool ranges are not confidence intervals.

For pooled controls, SD falls from .561 to .076 after integrating out sampling. These controls still condition on imperfect title classes and a single source capture per work; they do not calibrate artistic fidelity.

Model	11 common briefs		14 briefs	
	Pooled target	Class target	Equal family	Covariance
GPT Image 1	1.557	1.366	1.551	2.103
GPT Image 2	1.099	1.130	1.243	1.511
2.5 Flare	1.726	1.539	1.765	1.636
2.5 Sunburst	1.633	1.525	1.680	1.617
Nano Banana 2	0.987	0.948	1.203	1.385
FLUX.2 Max	0.750	0.786	0.808	0.928

Table 7: Descriptive error under alternative targets and metrics. The class target uses title-derived water, built and land strata; the three mixed briefs are excluded. Each column uses its own reference normalizer; absolute values across columns are not directly comparable. Covariance is fitted only on development works, with fixed 50% shrinkage.

C.2 Magnitude, artist coverage and coordinate weighting

The cross-repeat version of Equation 5 is exact for the retained vectors. Its variation term replaces squared deviations with cross-products between scene-centered repeats. The term can be negative in finite samples, although its independent-repeat population target is nonnegative. Scalar calibration fits only the other 13 scenes. All fold denominators Q are positive. GPT Image 2’s fitted scalars range from .438 to .459 and FLUX’s from .598 to .694; all fold-specific scores are retained. No calibration parameter is selected by minimizing its own held-out score. The subsequent deletion check refits the entire procedure on each remaining 13-scene panel, rather than treating overlapping folds as independent replications. Its mean GPT Image 2-minus-FLUX error differences range from −.189 to −.137.

Model	Squared contrast magnitude Q	Alignment ratio $\beta/\sqrt{Q}$
GPT Image 1	2.449	0.600
GPT Image 2	2.223	0.670
2.5 Flare	2.281	0.511
2.5 Sunburst	2.289	0.545
Nano Banana 2	0.994	0.444
FLUX.2 Max	0.741	0.546

Table 8: Descriptive magnitude and alignment of painter contrasts. $Q = 1$ equals the reference squared size, and $D = 1 - 2\beta + Q$. The alignment ratio can fall outside $[-1, 1]$ in finite samples.

For artist pairs, both generated and reference contrasts are direct differences between the two artists; the denominator is the squared reference-pair distance. Leaving one artist out recenters the remaining three generated and reference conditions. Reference singular components are the three rank-one terms of the centered four-by-31 reference matrix: each combines an artist contrast with a feature direction, rather than representing one artist. Component aligned responses project onto each term separately.

Model	Component aligned response			Aligned response after omitting			
	First	Second	Third	Monet	Sisley	Pissarro	Cézanne
GPT Image 1	1.248	0.310	0.356	1.140	1.072	0.840	0.389
GPT Image 2	1.156	0.617	0.800	1.195	0.972	0.992	0.651
2.5 Flare	1.022	0.211	0.383	1.034	0.734	0.802	0.222
2.5 Sunburst	1.082	0.250	0.417	1.116	0.840	0.762	0.284
Nano Banana 2	0.513	0.371	0.195	0.569	0.446	0.388	0.276
FLUX.2 Max	0.666	0.021	0.183	0.539	0.511	0.527	0.096

Table 9: Descriptive artist coverage of the aligned response. Reference components account for 66.3%, 20.6% and 13.0% of centroid variation. Each omission recomputes the target and recenters the other three artists.

Table 9 shows that aggregate positive alignment is uneven across those components.

Equal-family weighting assigns each coordinate inverse family size (11, 8 or 12). The covariance sensitivity uses 221 development works with equal painter mass. For their weighted covariance Σ , the metric is $[\frac{1}{2}\Sigma + \frac{1}{2}\text{tr}(\Sigma)I/31]^{-1}$, with fixed shrinkage and no tuning on evaluated images. These weighting choices are not independently validated artistic metrics.

FLUX retains the lowest point estimate under all 31 single-coordinate deletions; the report retains every coordinate contribution, including negative terms. Without texture, Flare and Sunburst still have error estimates above the no-contrast value of one (1.580 and 1.604).

C.3 A design-aware energy correction

The primary energy values use empirical distributions, including their zero diagonal distances. To estimate energy between a fixed empirical reference and the equally weighted mixture of scene-conditional output distributions, cross-scene generated pairs already have the required independent draws. Only the same-scene generated term needs adjustment. With two repeats and S scenes, the descriptive correction is

$$\hat{\mathcal{E}}_{\text{strata}} = \hat{\mathcal{E}}_{\text{empirical}} - \frac{1}{2S^2} \sum_s \|y_{s1} - y_{s2}\|.$$

The reference within-distance remains empirical because this target conditions on the finite reference panel. A pooled IID $n/(n-1)$ multiplier would also alter cross-scene terms and estimate

a different target. This correction retains the 21 negative named-minus-generic differences. It does not solve unequal historical and generated content mixtures, and no new energy tests are performed.

C.4 Uncertainty and synthetic coverage checks

The Student procedure uses 14 scene summaries. For independent scene errors with fixed conditional means δ_s , its estimated variance has expectation

$$\mathbb{E}[s_{\delta}^2/S] = \text{Var}(\bar{\delta}) + \frac{1}{S(S-1)} \sum_s (\delta_s - \bar{\delta})^2.$$

It therefore includes real between-scene heterogeneity as well as request noise. This explains why it is not a direct estimate of fresh-request variance for exactly the fixed panel. It also does not provide an exact coverage guarantee: non-Gaussian cross-products and dependence can affect the Student approximation.

The initial 5,000-run scenarios (seed 2026091202) use the six configurations, 14 scenes, two repeats and 21-endpoint family, with unit reference magnitude. Gaussian, variance-standardized t_3 /heteroskedastic and fixed scene-interaction errors give simultaneous coverage 95.96%, 97.42% and 99.68%. Their independent centered noise power is .5. Allocating half the noise to a model-specific state shared by all scenes and both repeats reduces coverage to 0.04%, with mean corrected-error bias near .25.

A subsequent sweep uses independent Gaussian errors at powers .5, 1.5 and 3, with either isotropic or rank-three covariance aligned with the reference components (5,000 runs per case). Coverage ranges from 95.06% to 96.80%, with Monte Carlo SE .25–.31 percentage points. For comparison, directly centered repeat differences estimate observed powers of .424–2.596 relative to H . These stress models probe coverage approximations, not actual service independence; no significance threshold is selected from them.

C.5 Source-quality sensitivity

AI assistants inspect all 649 reference and 221 development reproductions using contact sheets and enlarged peripheral regions, recording removable calibration targets, frames, margins and labels. Uncertain boundaries remain unchanged. Reference content is visually classified before comparison with the title label; mixed or unclear cases retain that label.

The audit selects crops for 90 reference and 41 development images, including ten calibration-strip cases; 43 uncertain boundaries remain unchanged. Visual classes differ from 230 title assignments, while 42 mixed or unclear cases retain their original labels. These AI-coded sensitivities did not undergo human verification and are not independent ground truth.

Audited rectangular crops follow the original orientation and ICC rules, before aspect-preserving Lanczos resizing to a 512-pixel short side without upsampling. All 31 features and all works are retained. We compare corrected references with original scaling, then refit development IQR scaling and consistently transform generated vectors. Label sensitivity uses the same 11 briefs. Versioned records preserve audit decisions, new vectors and original results; model scores do not select corrections. All six aligned-response intervals remain above zero with either scaler. With original scaling, corrected regions give FLUX $D = 0.832$. After refitting, (uncalibrated, held-out calibrated) errors are: GPT Image 1 (1.631, 0.693), GPT Image 2 (1.259, 0.579), Flare (1.737, 0.760), Sunburst (1.577, 0.711), Nano Banana 2 (1.121, 0.846) and FLUX (0.872, 0.762). FLUX and GPT Image 2 retain the respective minima.

D Image selection and provenance

The generated examples use the first declared brief and first repeat, selected after analysis without appearance- or score-based curation. The brief reads: “A broad river bends past a gravel bank. Three tall trees stand on the far bank, with a low hill behind them under a pale overcast sky.” Every prompt ends with “No text or frame.” Only the clauses in Section 3 change.

References in Figure 1 are the lexicographically first measured work IDs per painter classified as water-organized in the completed audit, with recorded public-domain status. The Sisley and Pissarro reproductions use audited crops removing calibration strips; Monet and Cézanne retain their full regions. The manifest records image identities, provenance, crop boxes and exact prompts. The paintings are not matched to the generated brief.

E Complete model and painter summaries

Tables 10–11 and Figures 4–5 summarize model comparisons, error components and collection-level proximity and spread. *Energy discrepancy* combines between-collection distances and internal spread; lower values mean greater proximity (Appendices B and C). Naming lowers energy relative to the generic clause in 21 of 24 model–painter cells, including the correction for repeated scenes. All 24 named collections have smaller total feature variance than their historical counterparts. These descriptive comparisons use 28 named or generic outputs against the recorded reference panel, with different subject mixtures; they do not establish agreement of painter contrasts.

Model A	Model B	$D_A - D_B$ [simultaneous interval]
GPT Image 1	GPT Image 2	0.346 [−0.400, 1.093]
GPT Image 1	2.5 Flare	−0.165 [−1.027, 0.697]
GPT Image 1	2.5 Sunburst	−0.069 [−1.181, 1.043]
GPT Image 1	Nano Banana 2	0.463 [−0.602, 1.528]
GPT Image 1	FLUX.2 Max	0.771 [−0.037, 1.580]
GPT Image 2	2.5 Flare	−0.512 [−1.150, 0.126]
GPT Image 2	2.5 Sunburst	−0.415 [−1.181, 0.350]
GPT Image 2	Nano Banana 2	0.116 [−0.922, 1.155]
GPT Image 2	FLUX.2 Max	0.425 [−0.161, 1.011]
2.5 Flare	2.5 Sunburst	0.096 [−0.286, 0.478]
2.5 Flare	Nano Banana 2	0.628 [−0.283, 1.540]
2.5 Flare	FLUX.2 Max	0.937 [0.256, 1.618]
2.5 Sunburst	Nano Banana 2	0.532 [−0.436, 1.500]
2.5 Sunburst	FLUX.2 Max	0.840 [0.058, 1.623]
Nano Banana 2	FLUX.2 Max	0.309 [−0.847, 1.464]

Table 10: All 15 paired model comparisons in the prespecified inferential family. Intervals are approximate 95% simultaneous intervals adjusted across all 21 endpoints. A negative contrast favors model A on the stated feature-geometry target.

Model	Measure	Monet	Sisley	Pissarro	Cézanne
GPT Image 1	Energy change	−2.515	−2.739	−2.628	−1.336
	Variance ratio	0.479	0.694	0.603	0.598
GPT Image 2	Energy change	−0.602	−1.092	−0.195	−0.373
	Variance ratio	0.283	0.312	0.303	0.303
2.5 Flare	Energy change	0.962	−0.008	0.429	−1.013
	Variance ratio	0.269	0.371	0.321	0.275
2.5 Sunburst	Energy change	0.267	−0.718	−0.091	−1.653
	Variance ratio	0.248	0.274	0.273	0.318
Nano Banana 2	Energy change	−0.513	−1.381	−0.741	−1.102
	Variance ratio	0.829	0.708	0.572	0.877
FLUX.2 Max	Energy change	−0.707	−0.886	−1.080	−1.431
	Variance ratio	0.340	0.416	0.570	0.622

Table 11: Complementary empirical distribution summaries. Energy change is named-minus-generic reference energy; variance ratios are generated-to-reference. Negative changes favor naming; ratios below one indicate narrower generated feature distributions.

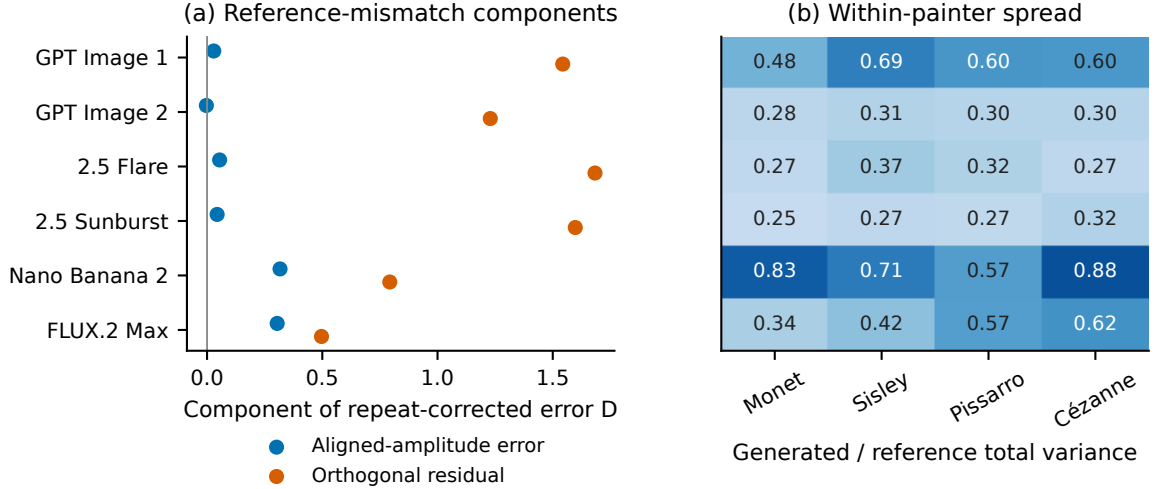


Figure 4: Descriptive summaries for the original reference target. (a) Amplitude error and the orthogonal residual—differences perpendicular to the overall labeled reference pattern—sum to D (Equation 4). (b) Generated/reference variance ratios for every model and painter. Total spread includes variation across fixed briefs and is not a mode-collapse measure.

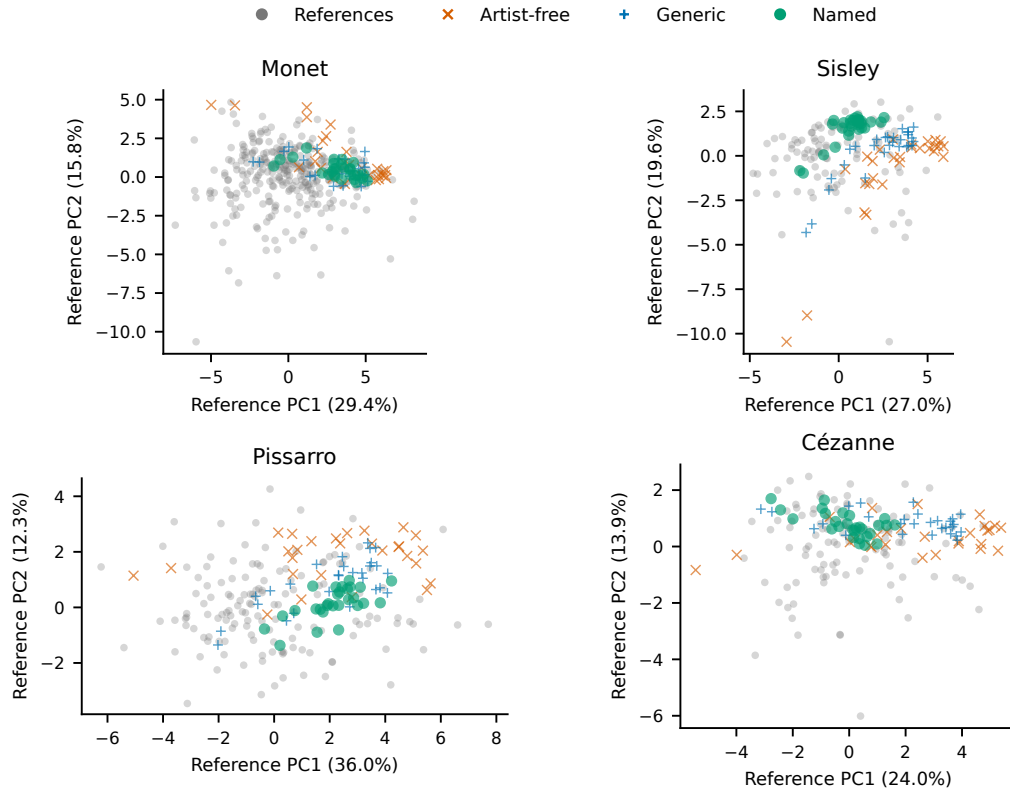


Figure 5: GPT Image 2.5 Sunburst’s named outputs appear more concentrated than the reference works in all four projections. This model illustration was selected before the primary feature outcomes were inspected. Each panel uses principal components fitted only to that painter’s reference works, so axes differ between painters. The 14 generated briefs and two repeats have a different content distribution from the historical panel. Projections are descriptive; numerical summaries use all 31 coordinates.

F Supporting interventions

These earlier cohorts address related but different questions (Table 12). They are not pooled with the six-model experiment or treated as independent-investigator replications. Historical model labels identify recorded request configurations, not verified underlying models (Appendix G).

Cohort	Comparison	Role in this paper
Four-painter exploration	649 references; 1,920 generated outputs	Describes reference/generated separation and motivates the controlled test.
Controlled naming	1,006 outputs; 38 Monet and 32 Cézanne references	Six named/free and two detailed/short prompt contrasts; subsequent transformation and conditional-variance diagnostics.
Palette interventions	Two separate 192-output collections	Named-minus-generic chroma response; no pooling across collections.
Generic-clause comparison	96 Cézanne/generic outputs on 24 new scenes	Tests a proximity gain beyond a generic painting clause.
Fixed-map transfer	240 FLUX/Cézanne outputs on 12 new scenes	Prospective counterexample to the historical opposing energy/prediction ordering.

Table 12: Supporting evidence is kept separate from the new six-model experiment. Different prompts, panels and finite-sample targets do not form one pooled test.

F.1 Proximity and spread under naming

The four-painter exploration used 1,536 named and 384 artist-free outputs, three prompt methods and 16 scenes. Its 24 model–painter–method cells had total variance ratios .206–.376 and grouped generated/reference discrimination accuracy .940–.983. Pooled named energy medians for Monet, Sisley, Pissarro and Cézanne were 2.068, 1.678, 1.975 and 2.142; corresponding artist-free medians were 1.732, 1.878, 1.854 and 1.657. Only Sisley had a lower pooled named median. Matched-size real/real medians were .193, .215, .164 and .199. These post-result summaries describe that exploration, not a general naming effect.

The later controlled panel selected Monet and Cézanne after exploration. It crossed 24 scenes with three models, two clauses and three repeats; 142 short-scene outputs supplied two prompt-length comparisons. Reference panels contained 38 Monet and 32 Cézanne works, with water/built/land masses based on maintainer-run LLM visual coding. Named-minus-free energy changes were $-.846, -1.818$ for Nano Banana 2, $-.625, -.896$ for FLUX, and $-.234, -.056$ for GPT Image 2 (Monet, Cézanne). The first four null hypotheses were rejected in the original family of eight. A separate 72-output FLUX collection repeated both directions ($-.695, -.952$) with rejection in its own four-endpoint family.

A standalone 96-output GPT Image 2/Cézanne comparison on 24 new scenes reduced energy relative to the generic clause by 1.195 ($p = .00001$, fixed $\alpha = .025$), with named/generic total variance ratio .552. It randomized clauses within 48 two-position blocks, used reference class masses (3,11,18)/32, and tested a sharp no-effect null with 99,999 within-block sign draws and plus-one correction. This tests clause effects, not equality of energy alone. A generic clause therefore did not reproduce the full proximity gain in that panel, but proximity does not establish correct four-artist contrasts. Likewise, lower total spread does not establish loss of scene information: controlled FLUX/Monet held-repetition scene retrieval rose from 44.4% to 59.7% under naming. This measures relative feature distinguishability, not verified prompt adherence.

F.2 Palette responsiveness and a contrary transfer result

Both 192-output palette collections requested the historical GPT Image 2 alias, using six scenes, four repeats, muted/vivid instructions and free, generic, Monet and Cézanne clauses. Primary name-minus-generic chroma-response interactions were unresolved (Monet, Cézanne): $-.284, -.018$ in the first collection and $-.159, .037$ in the later one. Their adjusted intervals spanned zero in the original two-interaction family and the later four-endpoint family, which also included two FLUX naming contrasts. Secondary generic-minus-free responses were $-.853$ and $-.891$, with nominal 95% intervals excluding zero. Thus generic painting can attenuate measured palette response without an identified additional name effect. Two palette levels do not distinguish reduced gain from saturation or an operating-range shift, and unresolved interactions do not demonstrate equivalence.

Earlier maps translated and optionally isotropically rescaled artist-free feature vectors to predict named-prompt scene means, with fits based only on generated images and evaluation on held-out scenes. Actual naming had lower reference energy than translation plus isotropic scaling in all six model-painter averages. Adding scale improved named-mean prediction in all six while worsening reference energy in five. That ordering did not transfer to a prospective 240-output FLUX/Cézanne comparison with 12 new scenes and ten repeats, using the original fits unchanged. Translation/scaling improved both energy and repeat-corrected prediction error relative to translation: $-.390 [-.509, -.271]$ and $-5.047 [-7.880, -2.214]$, with approximate simultaneous jackknife intervals. The achieved prediction-error half-width exceeded its planning criterion. This contrary result is retained without extra sampling; it rules out treating the historical opposing ordering as a universal mechanism. These named/free maps differ from the retrospective calibration of painter contrasts in Table 4.

G Collection provenance and computational scope

Historical GPT Image 1 and GPT Image 2 labels identify the requested model names. A later technical negative control showed that the earlier image interface also accepted an invented image-model identifier. This does not show that valid names select the same underlying model, but successful request acceptance alone cannot authenticate their separation. Historical contrasts are therefore interpreted as recorded request-group comparisons. The current experiment uses explicit documented model identifiers and fixed routes; it does not reuse historical images as verified examples of the new variants. All current outputs decode to the requested square dimensions, but their response bodies do not echo model or quality fields. Labels therefore denote the documented request configurations, without independent checkpoint attestation. Underlying checkpoints and training corpora are not available for inspection.

The present experiment excludes technical probes and an earlier attempt closed before feature measurement because model selection could not be qualified. No image is shared between these collections. The earlier incomplete generic-clause cohort is likewise not pooled with its complete 96-image successor. Exact assignments, attempt outcomes and measurement exclusions are retained in the versioned manifests; selection was not based on visual success.